Aleksandar Fontana

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Education

PhD in **AI and Cybersecurity** — *Scuola Superiore Sant'Anna* — 2024 – present

Master in Physics of Complex Systems and Big Data — *University of Tor Vergata* — 2021 – 2024

Thesis: "Exploring the frontier of chatbot technology"

Bachelor in Physics — *University of Tor Vergata* — 2018 – 2021

Thesis: "Quantum computing and cosmology of the early universe"

Work Experience

AI Consultant (Netgroup S.p.A., Naples) — 2025 - present

Engineered an agent, Transformer-based Foundation Model, for cyber attack prediction from scratch.

AI/NLP Research Intern (Serco Italia S.p.A, Rome) — 2023 - 2024

Engineered a Small LLM implementing BitNet 1.58-bit quantization for efficient deployment.

Head of Board of NPO (Element Gaming) — 2016 - present

Led community operations, coordinating a network of 100+ affiliated content creators.

Skills

LLM Alignment	Reinforcement Learning (PPO, GRPO, GTPO), TRL, Hugging Face
AI for Cybersecurity	Vulnerability Detection, Threat Intelligence, Custom Transformers, Tabular Data Modeling
Core ML	Python, PyTorch, CUDA, Git, Qiskit
Soft Skills	Team Leadership, Community Management, Event Coordination, Effective Communication
Languages	**Italian**: Native **English**: Professional Proficiency

Publications

On the Hidden Objective Biases of Group-based Reinforcement Learning — In ACL 2026

Developed a **unified surrogate formulation** to analyze **GRPO**, revealing theoretical structural mismatches and key limitations like prefix token gradient biases.

GTPO: Stabilizing Group Relative Policy Optimization via Gradient and Entropy Control

Code Repository — Submitted to TACL 2026

Designed a KL-free policy optimization algorithm mitigating post training LLM policy collapse. Boosted reasoning performance by up to **15%** on OOD benchmarks (AIME, AMC) vs. GRPO.

Improving LLM Reasoning for Vulnerability Detection via Group Relative Policy Optimization

Code Repository — 2025

Applied **GRPO** to enhance LLM reasoning for vulnerability detection, achieving up to **+29 macro** F_1 and **+17 accuracy points** over SFT, and utilized an LLM-as-a-Judge (LLMaaJ) to evaluate the usefulness of the answers.

TITAN: Graph-Executable Reasoning for Cyber Threat Intelligence Code Repository — 2025

Developed a framework mapping natural-language cyber queries to executable reasoning paths over a structured MITRE-based knowledge graph.

Unmasking model behavior: How LLMs reason on vulnerability detection — In ARES 2025